IMPORTS

```
!pip install tensorflow
```

```
Collecting tensorflow
  Downloading tensorflow-2.21.0-cp312-cp312-
manylinux_2_27_x86_64.whl.metadata (4.4 kB)
Requirement already satisfied: absl-py>=1.0.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (1.4.0)
Collecting astunparse>=1.6.0 (from tensorflow)
  Downloading astunparse-1.6.3-py2.py3-none-any.whl.metadata (4.4 kB)
Collecting flatbuffers>=25.9.23 (from tensorflow)
  Downloading flatbuffers-25.12.19-py2.py3-none-any.whl.metadata (1.0
kB)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1
in /usr/local/lib/python3.12/dist-packages (from tensorflow) (0.7.0)
Collecting google_pasta>=0.1.1 (from tensorflow)
  Downloading google_pasta-0.2.0-py3-none-any.whl.metadata (814 bytes)
Collecting libclang>=13.0.0 (from tensorflow)
  Downloading libclang-18.1.1-py2.py3-none-
manylinux2010_x86_64.whl.metadata (5.2 kB)
Requirement already satisfied: opt_einsum>=2.3.2 in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (26.0)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (6.33.5)
Requirement already satisfied: requests<3,>=2.21.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (2.32.4)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (75.2.0)
Requirement already satisfied: six>=1.12.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (2.1.2)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (1.78.0)
Requirement already satisfied: keras>=3.12.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (3.13.2)
Requirement already satisfied: numpy>=1.26.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow) (2.0.2)
Collecting h5py<3.15.0,>=3.11.0 (from tensorflow)
  Downloading h5py-3.14.0-cp312-cp312-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (2.7 kB)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in
```

```
google_pasta-0.2.0 h5py-3.14.0 libclang-18.1.1 tensorflow-2.21.0
wheel-0.46.3
```

```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image,ImageFilter,ImageEnhance # lib for the image I/O
and basic transforms
import os
import math
import json
import glob
import time
import random   # generating the random no.s
from pathlib import Path  # good way to handle the file sytem
from collections import Counter  # dictionary for counting the
occurence of each time
from tensorflow import keras
from tensorflow.keras import Sequential
from tensorflow.keras.utils import register_keras_serializable
from tensorflow.keras.layers import
Conv2D,BatchNormalization,ReLU,Add,Layer,Dropout,MultiHeadAttention,La
yerNormalization,Dense,Lambda,MaxPooling2D
```

01—Dataset Preparation

```
# before running this command you must upload the zip folder fo the
dataset from my github:- https://github.com/iAdityaDev/Optical-
Character-Recognition.git
!unzip dataset.zip
```

```
Archive:  dataset.zip
   creating: dataset/
   creating: dataset/images/
   creating: dataset/images/train/
  inflating: dataset/images/train/image_001.jpg
  inflating: dataset/images/train/image_002.jpg
  inflating: dataset/images/train/image_003.jpg
  inflating: dataset/images/train/image_004.jpg
  inflating: dataset/images/train/image_005.jpg
  inflating: dataset/images/train/image_006.jpg
  inflating: dataset/images/train/image_007.jpg
  inflating: dataset/images/train/image_008.jpg
  inflating: dataset/images/train/image_009.jpg
  inflating: dataset/images/train/image_010.jpg
  inflating: dataset/images/train/image_011.jpg
  inflating: dataset/images/train/image_012.jpg
  inflating: dataset/images/train/image_013.jpg
```

```
  inflating: dataset/images/train/image_014.jpg
  inflating: dataset/images/train/image_015.jpg
  inflating: dataset/images/train/image_016.jpg
  inflating: dataset/images/train/image_017.jpg
  inflating: dataset/images/train/image_018.jpg
  inflating: dataset/images/train/image_019.jpg
  inflating: dataset/images/train/image_020.jpg
  inflating: dataset/images/train/image_021.jpg
  inflating: dataset/images/train/image_022.jpg
  inflating: dataset/images/train/image_023.jpg
  inflating: dataset/images/train/image_024.jpg
  inflating: dataset/images/train/image_025.jpg
  inflating: dataset/images/train/image_026.jpg
  inflating: dataset/images/train/image_027.jpg
  inflating: dataset/images/train/image_028.jpg
  inflating: dataset/images/train/image_029.jpg
  inflating: dataset/images/train/image_030.jpg
  inflating: dataset/images/train/image_031.jpg
  inflating: dataset/images/train/image_032.jpg
  inflating: dataset/images/train/image_033.jpg
  inflating: dataset/images/train/image_034.jpg
  inflating: dataset/images/train/image_035.jpg
  inflating: dataset/images/train/image_036.jpg
  inflating: dataset/images/train/image_037.jpg
  inflating: dataset/images/train/image_038.jpg
  inflating: dataset/images/train/image_039.jpg
  inflating: dataset/images/train/image_040.jpg
  inflating: dataset/images/train/image_041.jpg
  inflating: dataset/images/train/image_042.jpg
  inflating: dataset/images/train/image_043.jpg
  inflating: dataset/images/train/image_044.jpg
  inflating: dataset/images/train/image_045.jpg
  inflating: dataset/images/train/image_046.jpg
  inflating: dataset/images/train/image_047.jpg
  inflating: dataset/images/train/image_048.jpg
  inflating: dataset/images/train/image_049.jpg
  inflating: dataset/images/train/image_050.jpg
  inflating: dataset/images/train/image_051.jpg
  inflating: dataset/images/train/image_052.jpg
  inflating: dataset/images/train/image_053.jpg
  inflating: dataset/images/train/image_054.jpg
  inflating: dataset/images/train/image_055.jpg
  inflating: dataset/images/train/image_056.jpg
  inflating: dataset/images/train/image_057.jpg
  inflating: dataset/images/train/image_058.jpg
  inflating: dataset/images/train/image_059.jpg
  inflating: dataset/images/train/image_060.jpg
  inflating: dataset/images/train/image_061.jpg
  inflating: dataset/images/train/image_062.jpg
```

```
  inflating: dataset/labels/test/label_395.txt
  inflating: dataset/labels/test/label_396.txt
  inflating: dataset/labels/test/label_397.txt
  inflating: dataset/labels/test/label_398.txt
  inflating: dataset/labels/test/label_399.txt
  inflating: dataset/labels/test/label_400.txt
```

```
DATASET_FORMAT = 'folder'
TRAIN_IMAGE_DIR = '/content/dataset/images/train'
TRAIN_LABEL_DIR = '/content/dataset/labels/train'
VAL_IMAGE_DIR = '/content/dataset/images/val'
VAL_LABEL_DIR = '/content/dataset/labels/val'
TEST_IMAGE_DIR = '/content/dataset/images/test'
TEST_LABEL_DIR = '/content/dataset/labels/test'
VOCAB_SAVE_PATH = 'vocab.json'
IMG_HEIGHT = 32
```

LOAD DATASET

```
def load_folder(img_dir,lbl_dir):
  items = []
  img_dir = Path(img_dir)
  lbl_dir = Path(lbl_dir)
  img_paths = sorted(img_dir.glob('*.jpg'))
  lbl_paths = sorted(lbl_dir.glob('*.txt'))

  for img_path, lbl_path in zip(img_paths, lbl_paths):
      text = lbl_path.read_text(encoding='utf-8').strip()
      if text:
          items.append({'image': str(img_path), 'text': text})
  return items

train_items = load_folder(TRAIN_IMAGE_DIR,TRAIN_LABEL_DIR)
val_items = load_folder(VAL_IMAGE_DIR,VAL_LABEL_DIR)
test_items = load_folder(TEST_IMAGE_DIR,TEST_LABEL_DIR)
print(train_items)
print(val_items)
print(test_items)

print(type(train_items))
print(len(train_items))
print(len(val_items))
print(len(test_items))
```

```
[{'image': '/content/dataset/images/train/image_001.jpg', 'text':
'Al'}, {'image': '/content/dataset/images/train/image_002.jpg',
'text': 'INFINITAMENTE AMABLE'}, {'image':
'/content/dataset/images/train/image_003.jpg', 'text': 'NIÑO JESUS.'},
```

```
'/content/dataset/images/test/image_388.jpg', 'text': 'Y lo cierto es,
que si sabe la muerte de su padre recono-'}, {'image':
'/content/dataset/images/test/image_389.jpg', 'text': 'ce el primer
lugar que tiene su madre para acusar, y las'}, {'image':
'/content/dataset/images/test/image_390.jpg', 'text': 'veras con que
lo haze, y lo fia todo de su deligencia: y el'}, {'image':
'/content/dataset/images/test/image_391.jpg', 'text': 'hijo esta tan
lexos de remitir, que es una de las causas'}, {'image':
'/content/dataset/images/test/image_392.jpg', 'text': 'que mas obliga
a su madre a insistir en el castigo por la'}, {'image':
'/content/dataset/images/test/image_393.jpg', 'text': 'mano de la
justicia, el temor no se le pierda este hijo'}, {'image':
'/content/dataset/images/test/image_394.jpg', 'text': 'varon, unico
sucessor en el mayorazdo de su padre, mo-'}, {'image':
'/content/dataset/images/test/image_395.jpg', 'text': 'ço alentado, y
con tantos agravios a los ojos.'}, {'image':
'/content/dataset/images/test/image_396.jpg', 'text': 'Y en quanto a
la hija, que primero represento tan'}, {'image':
'/content/dataset/images/test/image_397.jpg', 'text': 'encarecidamente
la muerte de su padre, y en casando-'}, {'image':
'/content/dataset/images/test/image_398.jpg', 'text': 'se con el
matador solicita su causa, no es de momento;'}, {'image':
'/content/dataset/images/test/image_399.jpg', 'text': 'tam, por no
poder concurrir con la viuda, como por'}, {'image':
'/content/dataset/images/test/image_400.jpg', 'text': 'hallarse
subdita de su marido, y prohibida a acusalle: y'}]
<class 'list'>
380
73
20
```

Visualize Images

```
samples = random.sample(train_items, min(18, len(train_items)))
fig, axes = plt.subplots(len(samples), 1, figsize=(14, len(samples) *
1.4))
print(axes)
if len(samples) == 1:
    axes = [axes]

for ax, item in zip(axes, samples):
    img = Image.open(item['image']).convert('L')
    ax.imshow(img, cmap='gray', aspect='auto')
    ax.set_title(f"{item['text']}   [{img.size[0]}x{img.size[1]}px]",
                 fontsize=10, loc='left')
    ax.axis('off')
```

```
plt.tight_layout()
plt.show()
```

```
[<Axes: > <Axes: > <Axes: > <Axes: > <Axes: > <Axes: > <Axes: > <Axes:
>
 <Axes: > <Axes: > <Axes: > <Axes: > <Axes: > <Axes: > <Axes: > <Axes:
>
 <Axes: > <Axes: >]
```

Adelantense en la virtud, como [381x29px]

Adelantenſe en la virtud , como

severancia con que ha seguido, y sigue este pleyto: y la [877x49px]

ſeuerancia con que ha ſeguido,y ſigue eſte pleyto : y la

cusationibus, resoluit Flores de Mena d. lib. I. q. 7. nu. 5. [877x49px]

cuſationibus,reſoluit Flores de Mena d.lib.1.q.7. nu. 5.

mez Pimentel, ha recibido de Antonio Gomez su [813x48px]

mez Pimentel,ha recibido de Antonio Gomez ſu

daron, que el mandamiento despacha [573x47px]

,daron, que el mandamiẽto deſpacha

siendo menor otorgó en fauor de su padre dos car [772x51px]

ſiendo menor otorgò en fauor de ſu padre dos car

mentel, y Antonio Gomez su padre, por la qual consta [817x48px]

mẽtel,y AntonioGomez ſupadre,porla qual cõſta

dre de Iacinto Gomez, para que diesse bienes de su [776x48px]

dre de Iacinto Gomez,para que dieſſe bienes de ſu

las dos cartas de pago, que antes desta tenia otor- [774x55px]

las dos cartas de pago, que antes deſta tenia otor-

tir como Niño entre los Doctores, [492x44px]

tir como Niño entre los Doctores,

lo titulo es muy vuestra; y por [492x44px]

lo titulo · es muy vueſtra ; · y por

traslado dellas a las partes, y que Antonio Gomez [751x49px]

traſlado dellas a las partes, y que Antonio Gomez

los bienes de doña Beatriz de Silua, entre Atonio [800x49px]

los bienes de doña Beatriz de Silua , entre Atonio

o Niño tierno, y Dios Eterno, [381x33px]

ò Niño tierno , y Dios Eterno.

Character Vocabulary

```
# building this vocab cause model works on the mathematical values
only thats why making vocab only

char_counter = Counter()
for it in train_items:
    char_counter.update(it['text'])

print(f'20 mostappeared:')
for ch, cnt in char_counter.most_common(20):
    print(f'{repr(ch):8s} {cnt:,}')

# extra_chars = ['Q', '…']
# for ch in extra_chars:
#     char_counter[ch] = 1

val_chars = set()
for it in val_items:
    val_chars.update(it['text'])
unseen = val_chars - set(char_counter.keys())

for ch in unseen:
  char_counter[ch] = 1
unseen = {}
if unseen:
    print(f'{sorted(unseen)}')
else:
    print('unseen in training.')
```

```
20 mostappeared:
' '      2,391
'e'      1,558
'a'      1,296
'o'      1,043
's'      862
'i'      778
'r'      771
'n'      747
'd'      642
'l'      555
't'      538
'u'      492
'c'      435
','      290
'm'      290
'p'      286
'y'      180
'q'      157
'g'      126
```

```
'-'      105
unseen in training.
```

```
sorted_chars = sorted(char_counter.keys())
char2idx = {c: i+1 for  i,c in enumerate(sorted_chars)}
idx2char = {i + 1: c for i, c in enumerate(sorted_chars)}
idx2char[0] = '<blank>'
NUM_CLASSES = len(sorted_chars) + 1
print(NUM_CLASSES)
# print(f'Characters      : {repr("".join(sorted_chars))}')

test_text = train_items[0]['text']

encoded = []
for c in test_text:
  if c in char2idx:
    encoded.append(char2idx[c])
decoded   = ''.join(idx2char[i] for i in encoded)

print(idx2char)
```

```
77
{1: ' ', 2: '!', 3: '&', 4: '(', 5: ')', 6: ',', 7: '-', 8: '.', 9:
'0', 10: '1', 11: '2', 12: '3', 13: '4', 14: '5', 15: '6', 16: '7',
17: '8', 18: '9', 19: ':', 20: ';', 21: '?', 22: 'A', 23: 'B', 24:
'C', 25: 'D', 26: 'E', 27: 'F', 28: 'G', 29: 'H', 30: 'I', 31: 'J',
32: 'L', 33: 'M', 34: 'N', 35: 'O', 36: 'P', 37: 'Q', 38: 'R', 39:
'S', 40: 'T', 41: 'U', 42: 'V', 43: 'Y', 44: 'Z', 45: 'a', 46: 'b',
47: 'c', 48: 'd', 49: 'e', 50: 'f', 51: 'g', 52: 'h', 53: 'i', 54:
'j', 55: 'l', 56: 'm', 57: 'n', 58: 'o', 59: 'p', 60: 'q', 61: 'r',
62: 's', 63: 't', 64: 'u', 65: 'v', 66: 'x', 67: 'y', 68: 'z', 69:
'Ñ', 70: 'á', 71: 'ç', 72: 'é', 73: 'í', 74: 'ñ', 75: 'ó', 76: '…', 0:
'<blank>'}
```

```
sorted_chars = sorted(char_counter.keys())
char2idx = {c: i+1 for  i,c in enumerate(sorted_chars)}
idx2char = {i + 1: c for i, c in enumerate(sorted_chars)}
idx2char[0] = '<blank>'
NUM_CLASSES = len(sorted_chars) + 1
print(NUM_CLASSES)


test_text = train_items[0]['text']
encoded   = [char2idx[c] for c in test_text if c in char2idx]
decoded   = ''.join(idx2char[i] for i in encoded)
print(encoded)
print(decoded)
```

```
77
[22, 55]
Al
```

```
###saving this so that i can reuse it later without generating vocab
again again
vocab_data = {
    'chars'      : sorted_chars,
    'num_classes': NUM_CLASSES,
    'char2idx'   : char2idx,
    'note'       : 'Index 0 is CTC blank. Characters start at index
1.'
}

with open(VOCAB_SAVE_PATH, 'w', encoding='utf-8') as f:
    json.dump(vocab_data, f, ensure_ascii=False, indent=2)

def filter_items(items, char2idx):
    kept, dropped = [], []
    for it in items:
        encoded = [char2idx[c] for c in it['text'] if c in char2idx]
        if len(encoded) > 0:
            it['encoded'] = encoded
            kept.append(it)
        else:
            dropped.append(it)
    return kept, dropped

train_items, train_dropped = filter_items(train_items, char2idx)
val_items,   val_dropped   = filter_items(val_items,   char2idx)

print(f'{len(val_items)}')

####json files for train_items and val_itmes again so that i can give
use it later
with open('train_items.json', 'w') as f:
    json.dump(train_items, f ,  indent=2)
with open('val_items.json', 'w') as f:
    json.dump(val_items, f ,  indent=2)
```

```
73
```

02—Model Building

HyperParameters

```
IMG_HEIGHT   = 32
D_MODEL      = 64
NUM_HEADS    = 4
NUM_LAYERS   = 2
D_FF         = 128
DROPOUT      = 0.1
```

```
def conv_bn_relu(x,filters,kernel_size=3,strides=1,padding='same'):
  x =
Conv2D(filters,kernel_size,strides=strides,padding=padding,use_bias =
False)(x)
  x = BatchNormalization()(x)
  x = ReLU()(x)
  return x

def residual_block(x, filters, strides=1):
    shortcut = x

    out = Conv2D(filters, 3, strides=strides,
                         padding='same', use_bias=False)(x)
    out = BatchNormalization()(out)
    out = ReLU()(out)
    out = Conv2D(filters, 3, padding='same', use_bias=False)(out)
    out = BatchNormalization()(out)

    if strides != 1 or x.shape[-1] != filters:
        shortcut = Conv2D(filters, 1, strides=strides,
                                  use_bias=False)(shortcut)
        shortcut = BatchNormalization()(shortcut)

    out = Add()([out, shortcut])
    out = ReLU()(out)
    return out

def build_cnn_backbone(x, out_channels):
    x = conv_bn_relu(x, 64, 3)
    x = MaxPooling2D(2, padding='same')(x)

    x = residual_block(x, 128, strides=2)
    x = residual_block(x, 128, strides=1)

    x = residual_block(x, 256, strides=2)


    # x = residual_block(x, 128, strides=1)


    # x = Conv2D(out_channels, 8, strides=(3, 3),padding='same',
use_bias=False)(x)
    # x = BatchNormalization()(x)
    # x = ReLU()(x)


    x = residual_block(x, 256, strides=1)


    x = Conv2D(out_channels, 3, strides=(1, 1),padding='same',
use_bias=False)(x)
```

```
    x = BatchNormalization()(x)
    x = ReLU()(x)

    x = Conv2D(out_channels, 3, strides=(1, 1),padding='same',
use_bias=False)(x)
    x = BatchNormalization()(x)
    x = ReLU()(x)
    return x

print('CNN formed')
```

```
CNN formed
```

Positional Encoding

```
@register_keras_serializable()
class SinusoidalPositionalEncoding(Layer):
  def __init__(self,d_model,max_len=2048,dropout_rate=0.1,**kwargs):
    super().__init__(**kwargs)
    self.dropout = Dropout(dropout_rate)

    # encoding table in numpy
    pe = np.zeros((max_len,d_model),dtype=np.float32)
    pos = np.arange(max_len)[:,np.newaxis]


    div = np.exp(np.arange(0,d_model,2)*(-math.log(10000.0) /
d_model))
    pe[:, 0::2] = np.sin(pos*div)
    pe[:, 1::2] = np.cos(pos*div)
    self._pe = pe[np.newaxis]

  def call(self, x, training=False):
      T  = tf.shape(x)[1]
      pe = tf.constant(self._pe, dtype=tf.float32)


      x  = x + pe[:, :T, :]
      return self.dropout(x, training=training)
```

Transformer Encoder

```
@register_keras_serializable()
class TransformerEncoderLayer(Layer):
  def __init__(self,d_model,num_heads,d_ff,dropout_rate=0.1,**kwargs):
    super().__init__(**kwargs)
    self.attn =
MultiHeadAttention(num_heads=num_heads,key_dim=d_model//num_heads,drop
out=dropout_rate)
    self.ffn = Sequential([
```

```
        Dense(d_ff,activation='relu'),
        Dropout(dropout_rate),
        Dense(d_model)
    ])
    self.norm1 = LayerNormalization(epsilon=1e-6)
    self.norm2 = LayerNormalization(epsilon=1e-6)
    self.drop1 = Dropout( dropout_rate)
    self.drop2 = Dropout( dropout_rate)

  def call(self,x,training=False):
    normed = self.norm1(x)
    attn_out = self.attn(normed,normed,training=training)
    #sub layer feed forward
    x = x +
self.drop2(self.ffn(self.norm2(x),training=training),training=training
)
    return x

@register_keras_serializable()
class TransformerEncoder(Layer):
  ###stack on may encoder layers
  def
__init__(self,d_model,num_heads,num_layers,d_ff,dropout_rate=0.1,**kwa
rgs):
     super().__init__(**kwargs)
     self.layers_ = [

TransformerEncoderLayer(d_model,num_heads,d_ff,dropout_rate,name=f'enc
_{i}')
         for i in range(num_layers)
     ]
     self.norm = LayerNormalization(epsilon=1e-6)
  def call(self,x,training=False):
    for layer in self.layers_ :
      x = layer(x,training=training)
    return self.norm(x)

@register_keras_serializable()
class PoolHeight(keras.layers.Layer):
    def call(self,t):
        return tf.reduce_mean(t, axis=1)

    def compute_output_shape(self, input_shape):
        # (B, H, W, C) -> (B, W, C)
        return (input_shape[0], input_shape[2])

@register_keras_serializable()
class LogSoftmax(keras.layers.Layer):
    def call(self,t):
        return tf.nn.log_softmax(t, axis=-1)
```

```
    def compute_output_shape(self, input_shape):
        return input_shape

def build_ocr_model(num_classes, img_height=32,
d_model=256,num_heads=4, num_layers=3, d_ff=512, dropout=0.1):

    images = keras.Input(shape=(img_height, None, 1), name='images')

    feat = build_cnn_backbone(images, out_channels=d_model)


    feat = PoolHeight(name='pool_height')(feat)

    feat = SinusoidalPositionalEncoding(d_model, dropout_rate=dropout,
name='pos_enc')(feat)

    feat = TransformerEncoder(d_model, num_heads, num_layers,
d_ff,dropout_rate=dropout, name='transformer')(feat)

    logits    = Dense(num_classes, name='logits')(feat)

    log_probs = LogSoftmax(name='log_probs')(logits)
    return keras.Model(inputs=images, outputs=log_probs,
name='OCRModel')


model = build_ocr_model(
    num_classes = NUM_CLASSES,
    img_height  = IMG_HEIGHT,
    d_model     = D_MODEL,
    num_heads   = NUM_HEADS,
    num_layers  = NUM_LAYERS,
    d_ff        = D_FF,
    dropout     = DROPOUT
)

model.summary(line_length=80)
```

```
Model: "OCRModel"

┏━━━
```

```
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ conv2d (Conv2D)          │ (None, 32, None,     │          576 │ images[0]
[0]        │
│                          │ 64)                  │              │
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ batch_normalization      │ (None, 32, None,     │          256 │ conv2d[0]
[0]        │
│ (BatchNormalization)     │ 64)                  │              │
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ re_lu (ReLU)             │ (None, 32, None,     │            0 │
batch_normalizati… │
│                          │ 64)                  │              │
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ max_pooling2d            │ (None, 16, None,     │            0 │ re_lu[0]
[0]         │
│ (MaxPooling2D)           │ 64)                  │              │
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ conv2d_1 (Conv2D)        │ (None, 8, None,      │       73,728 │
max_pooling2d[0][… │
│                          │ 128)                 │              │
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ batch_normalization_1    │ (None, 8, None,      │          512 │
conv2d_1[0][0]     │
│ (BatchNormalization)     │ 128)                 │              │
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ re_lu_1 (ReLU)           │ (None, 8, None,      │            0 │
batch_normalizati… │
│                          │ 128)                 │              │
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ conv2d_2 (Conv2D)        │ (None, 8, None,      │      147,456 │ re_lu_1[0]
[0]       │
│                          │ 128)                 │              │
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
```

```
──────────┤
│ conv2d_3 (Conv2D)      │ (None, 8, None,    │        8,192 │
max_pooling2d[0][… │
│                        │ 128)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼──────────
──────────┤
│ batch_normalization_2 │ (None, 8, None,    │          512 │
conv2d_2[0][0]     │
│ (BatchNormalization)   │ 128)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼──────────
──────────┤
│ batch_normalization_3 │ (None, 8, None,    │          512 │
conv2d_3[0][0]     │
│ (BatchNormalization)   │ 128)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼──────────
──────────┤
│ add (Add)              │ (None, 8, None,    │            0 │
batch_normalizati… │
│                        │ 128)               │              │
batch_normalizati… │
├────────────────────────┼────────────────────┼──────────────┼──────────
──────────┤
│ re_lu_2 (ReLU)         │ (None, 8, None,    │            0 │ add[0][0]
│
│                        │ 128)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼──────────
──────────┤
│ conv2d_4 (Conv2D)      │ (None, 8, None,    │      147,456 │ re_lu_2[0]
[0]      │
│                        │ 128)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼──────────
──────────┤
│ batch_normalization_4 │ (None, 8, None,    │          512 │
conv2d_4[0][0]     │
│ (BatchNormalization)   │ 128)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼──────────
──────────┤
│ re_lu_3 (ReLU)         │ (None, 8, None,    │            0 │
batch_normalizati… │
│                        │ 128)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼──────────
──────────┤
```

```
│ conv2d_5 (Conv2D)      │ (None, 8, None,    │      147,456 │ re_lu_3[0]
[0]      │
│                        │ 128)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ batch_normalization_5 │ (None, 8, None,    │          512 │
conv2d_5[0][0]     │
│ (BatchNormalization)   │ 128)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ add_1 (Add)            │ (None, 8, None,    │            0 │
batch_normalizati… │
│                        │ 128)               │              │ re_lu_2[0]
[0]      │
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ re_lu_4 (ReLU)         │ (None, 8, None,    │            0 │ add_1[0]
[0]        │
│                        │ 128)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ conv2d_6 (Conv2D)      │ (None, 4, None,    │      294,912 │ re_lu_4[0]
[0]      │
│                        │ 256)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ batch_normalization_6 │ (None, 4, None,    │        1,024 │
conv2d_6[0][0]     │
│ (BatchNormalization)   │ 256)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ re_lu_5 (ReLU)         │ (None, 4, None,    │            0 │
batch_normalizati… │
│                        │ 256)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ conv2d_7 (Conv2D)      │ (None, 4, None,    │      589,824 │ re_lu_5[0]
[0]      │
│                        │ 256)               │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ conv2d_8 (Conv2D)      │ (None, 4, None,    │       32,768 │ re_lu_4[0]
```

```
[0]      |
|                         | 256)               |              |
|
|──────────────────────────┼────────────────────┼──────────────┼──────────────
──────────┤
| batch_normalization_7 | (None, 4, None,   |       1,024 |
conv2d_7[0][0]     |
| (BatchNormalization)  | 256)               |              |
|
|──────────────────────────┼────────────────────┼──────────────┼──────────────
──────────┤
| batch_normalization_8 | (None, 4, None,   |       1,024 |
conv2d_8[0][0]     |
| (BatchNormalization)  | 256)               |              |
|
|──────────────────────────┼────────────────────┼──────────────┼──────────────
──────────┤
| add_2 (Add)           | (None, 4, None,   |           0 |
batch_normalizati… |
|                         | 256)               |              |
batch_normalizati… |
|──────────────────────────┼────────────────────┼──────────────┼──────────────
──────────┤
| re_lu_6 (ReLU)        | (None, 4, None,   |           0 | add_2[0]
[0]        |
|                         | 256)               |              |
|
|──────────────────────────┼────────────────────┼──────────────┼──────────────
──────────┤
| conv2d_9 (Conv2D)     | (None, 4, None,   |     589,824 | re_lu_6[0]
[0]      |
|                         | 256)               |              |
|
|──────────────────────────┼────────────────────┼──────────────┼──────────────
──────────┤
| batch_normalization_9 | (None, 4, None,   |       1,024 |
conv2d_9[0][0]     |
| (BatchNormalization)  | 256)               |              |
|
|──────────────────────────┼────────────────────┼──────────────┼──────────────
──────────┤
| re_lu_7 (ReLU)        | (None, 4, None,   |           0 |
batch_normalizati… |
|                         | 256)               |              |
|
|──────────────────────────┼────────────────────┼──────────────┼──────────────
──────────┤
| conv2d_10 (Conv2D)    | (None, 4, None,   |     589,824 | re_lu_7[0]
[0]      |
```

```
│                         │ 256)                │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ batch_normalization_… │ (None, 4, None,    │        1,024 │
conv2d_10[0][0]    │
│ (BatchNormalization)   │ 256)                │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ add_3 (Add)            │ (None, 4, None,    │            0 │
batch_normalizati… │
│                         │ 256)                │              │ re_lu_6[0]
[0]      │
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ re_lu_8 (ReLU)         │ (None, 4, None,    │            0 │ add_3[0]
[0]        │
│                         │ 256)                │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ conv2d_11 (Conv2D)     │ (None, 4, None,    │      147,456 │ re_lu_8[0]
[0]      │
│                         │ 64)                 │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ batch_normalization_… │ (None, 4, None,    │          256 │
conv2d_11[0][0]    │
│ (BatchNormalization)   │ 64)                 │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ re_lu_9 (ReLU)         │ (None, 4, None,    │            0 │
batch_normalizati… │
│                         │ 64)                 │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ conv2d_12 (Conv2D)     │ (None, 4, None,    │       36,864 │ re_lu_9[0]
[0]      │
│                         │ 64)                 │              │
│
├────────────────────────┼────────────────────┼──────────────┼────────────
──────────┤
│ batch_normalization_… │ (None, 4, None,    │          256 │
conv2d_12[0][0]    │
│ (BatchNormalization)   │ 64)                 │              │
```

```
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ re_lu_10 (ReLU)          │ (None, 4, None,      │            0 │
batch_normalizati… │
│                          │ 64)                  │              │
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ pool_height              │ (None, None)         │            0 │
re_lu_10[0][0]     │
│ (PoolHeight)             │                      │              │
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ pos_enc                  │ (1, None, 64)        │            0 │
pool_height[0][0]  │
│ (SinusoidalPositiona…    │                      │              │
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ transformer              │ (1, None, 64)        │       67,072 │ pos_enc[0]
[0]      │
│ (TransformerEncoder)     │                      │              │
│
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ logits (Dense)           │ (1, None, 77)        │        5,005 │
transformer[0][0]  │
├──────────────────────────┼──────────────────────┼──────────────┼──────────────
──────────┤
│ log_probs                │ (1, None, 77)        │            0 │ logits[0]
[0]       │
│ (LogSoftmax)             │                      │              │
│
└──────────────────────────┴──────────────────────┴──────────────┴──────────────
──────────┘

 Total params: 2,886,861 (11.01 MB)

 Trainable params: 2,882,637 (11.00 MB)

 Non-trainable params: 4,224 (16.50 KB)
```

```python
def ctc_greedy_decode(log_probs, idx2char, blank_idx=0):

    best = tf.argmax(log_probs, axis=-1).numpy()
    print(best.shape[0])
    results = []
    for b in range(best.shape[0]):
```

```
        path = best[b].tolist()
        # Collapse repeats
        collapsed = [path[0]]
        for tok in path[1:]:
            if tok != collapsed[-1]:
                collapsed.append(tok)
        # blanks removal
        decoded = [t for t in collapsed if t != blank_idx]
        results.append(''.join(idx2char.get(t, '?') for t in decoded))
    return results


dummy   = tf.zeros([1, IMG_HEIGHT, 320, 1])
out     = model(dummy, training=False)
decoded = ctc_greedy_decode(out, idx2char)

1
```

Training

```
EPOCHS       = 50
BATCH_SIZE   = 8
LEARNING_RATE = 1e-3
CHECKPOINT_DIR = 'checkpoints'
Path(CHECKPOINT_DIR).mkdir(exist_ok=True)

def augment(img):
    if random.random() < 0.15:
        img =
img.filter(ImageFilter.GaussianBlur(radius=random.uniform(0.3, 1.2)))
    if random.random() < 0.3:
        img =
ImageEnhance.Brightness(img).enhance(random.uniform(0.75, 1.3))
        img = ImageEnhance.Contrast(img).enhance(random.uniform(0.8,
1.25))
    if random.random() < 0.2:
        arr = np.array(img, dtype=np.float32)
        arr = np.clip(arr + np.random.randn(*arr.shape) *
random.uniform(2, 6),0, 255).astype(np.uint8)
        img = Image.fromarray(arr)
    return img

def load_and_preprocess(path, label, text):
    img = tf.io.read_file(path)
    img = tf.image.decode_jpeg(img, channels=1)
    new_w = tf.cast(tf.shape(img)[1], tf.float32) * IMG_HEIGHT /
tf.cast(tf.shape(img)[0], tf.float32)
    img = tf.image.resize(img, [IMG_HEIGHT, tf.cast(new_w, tf.int32)])
```

```
    img = tf.cast(img, tf.float32) / 127.5 - 1.0
    return img, label, text


def make_dataset(items, shuffle):
    paths  = [it['image'] for it in items]
    labels = [it['encoded'] for it in items]
    texts  = [it['text']    for it in items]

    # pats  = [it['image'] for it in items]
    # laels = [it['encoded'] for it in items]
    # texts  = [it['text']    for it in items]

    max_len = max(len(l) for l in labels)
    labels_padded = [l + [0] * (max_len - len(l)) for l in labels]

    path_ds  = tf.data.Dataset.from_tensor_slices(paths)
    label_ds = tf.data.Dataset.from_tensor_slices(
        tf.constant(labels_padded, dtype=tf.int32)
    )
    text_ds  = tf.data.Dataset.from_tensor_slices(texts)

    ds = tf.data.Dataset.zip((path_ds, label_ds, text_ds))

    if shuffle:
        ds = ds.shuffle(buffer_size=len(items),
reshuffle_each_iteration=True)

    return (
        ds
        .map(load_and_preprocess, num_parallel_calls=tf.data.AUTOTUNE)
        .padded_batch(
            BATCH_SIZE,
            padded_shapes=([IMG_HEIGHT, None, 1], [None], []),
            padding_values=(-1.0, 0, '')
        )
        # .prefetch(1)
    )

train_ds = make_dataset(train_items, shuffle=True)
val_ds   = make_dataset(val_items,   shuffle=False)

print(f'{len(train_items)} {len(train_items)/BATCH_SIZE}')
print(f'{len(val_items)} ')
```

```
380 47.5
73
```

CTC loss and optimizer

```
def compute_ctc_loss(log_probs, labels, images):
    B = tf.shape(log_probs)[0]
    T = tf.shape(log_probs)[1]

    seq_lengths = tf.fill([B], T)

    label_lengths = tf.cast(
        tf.math.count_nonzero(labels, axis=1), tf.int32
    )

    logits_tm = tf.transpose(tf.cast(log_probs, tf.float32), [1, 0,
2])

    loss = tf.nn.ctc_loss(
        labels            = tf.cast(labels, tf.int32),
        logits            = logits_tm,
        label_length      = label_lengths,
        logit_length      = seq_lengths,
        logits_time_major = True,
        blank_index       = 0,
    )
    return tf.reduce_mean(loss)

try:
    optimizer =
keras.optimizers.AdamW(learning_rate=LEARNING_RATE,weight_decay=1e-4,
clipnorm=5.0)
except AttributeError:
    try:
        optimizer = keras.optimizers.experimental.AdamW(
            learning_rate=LEARNING_RATE, weight_decay=1e-4,
clipnorm=5.0)
    except AttributeError:
        optimizer = keras.optimizers.Adam(learning_rate=LEARNING_RATE,
clipnorm=5.0)

@tf.function
def train_step(images, labels):
    with tf.GradientTape() as tape:
        log_probs = model(images, training=True)
        loss = compute_ctc_loss(log_probs, labels, images)
    grads = tape.gradient(loss, model.trainable_weights)
    grads = [g if g is not None else tf.zeros_like(w)
             for g, w in zip(grads, model.trainable_weights)]
    optimizer.apply_gradients(zip(grads, model.trainable_weights))
    return loss


def ctc_greedy_decode(log_probs):
    best = tf.argmax(log_probs, axis=-1).numpy()
```

```
    results = []
    for row in best:
        col = [row[0]]
        # for t in row[1:]:

        for t in row[1:]:
            if t != col[-1]: col.append(t)
        results.append(''.join(idx2char.get(int(t), '') for t in col
if t != 0))
    return results


def levenshtein(s1, s2):
    if s1 == s2: return 0
    if len(s1) < len(s2): s1, s2 = s2, s1
    prev = list(range(len(s2)+1))

    for c1 in s1:
        cur = [prev[0]+1]
        for j, c2 in enumerate(s2):
            cur.append(min(prev[j]+(0 if c1==c2 else 1), cur[-1]+1,
prev[j+1]+1))
        prev = cur
    return prev[-1]


def val_epoch():
    preds, gts, total_loss, n = [], [], 0.0, 0
    for batch in val_ds:
        lp   = model(batch[0], training=False)
        loss = compute_ctc_loss(lp, batch[1], batch[0])
        total_loss += float(loss); n += 1
        preds.extend(ctc_greedy_decode(lp))
        gts.extend([t.numpy().decode() for t in batch[2]])

    tot_dist = sum(levenshtein(g, p) for g, p in zip(gts, preds))
    tot_len  = sum(len(g) for g in gts)
    cer = tot_dist / max(tot_len, 1)
    return total_loss / max(n, 1), cer


history = {'train_loss': [], 'val_loss': [], 'cer': []}
best_cer = float('inf')

for epoch in range(1, EPOCHS + 1):
    t0, train_losses = time.time(), []

    for batch in train_ds:
        loss = train_step(batch[0], batch[1])
        train_losses.append(float(loss))
```

```
        del batch

    val_loss, cer = val_epoch()
    avg_loss = sum(train_losses) / max(len(train_losses), 1)
    elapsed  = time.time() - t0

    history['train_loss'].append(avg_loss)
    history['val_loss'].append(val_loss)
    history['cer'].append(cer)

    marker = ''
    if cer < best_cer:
        best_cer = cer
        model.save(f'{CHECKPOINT_DIR}/best.keras')
        marker = '  ← best'

    print(f'Epoch {epoch:03d}/{EPOCHS}'
          f'loss: {avg_loss:.2f}'
          f'val_loss: {val_loss:.4f}'
          f'CER: {cer:.4f}')
```

```
WARNING:tensorflow:5 out of the last 6 calls to <function train_step
at 0x7d1c6f2c53a0> triggered tf.function retracing. Tracing is
expensive and the excessive number of tracings could be due to (1)
creating @tf.function repeatedly in a loop, (2) passing tensors with
different shapes, (3) passing Python objects instead of tensors. For
(1), please define your @tf.function outside of the loop. For (2),
@tf.function has reduce_retracing=True option that can avoid
unnecessary retracing. For (3), please refer to
https://www.tensorflow.org/guide/function#controlling_retracing and
https://www.tensorflow.org/api_docs/python/tf/function for  more
details.
WARNING:tensorflow:6 out of the last 7 calls to <function train_step
at 0x7d1c6f2c53a0> triggered tf.function retracing. Tracing is
expensive and the excessive number of tracings could be due to (1)
creating @tf.function repeatedly in a loop, (2) passing tensors with
different shapes, (3) passing Python objects instead of tensors. For
(1), please define your @tf.function outside of the loop. For (2),
@tf.function has reduce_retracing=True option that can avoid
unnecessary retracing. For (3), please refer to https://
www.tensorflow.org/guide/function#controlling_retracing and https://
www.tensorflow.org/api_docs/python/tf/function for more details.
```

```
---------------------------------------------------------------------------
KeyboardInterrupt                         Traceback (most recent call
last)
```

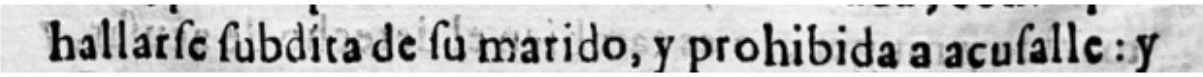
```
#         'LogSoftmax': LogSoftmax, ## earlier these classes were not
serialised that's imported them
#     },
#     safe_mode=False
# )

model = keras.models.load_model(
    '/content/best.keras',safe_mode=False
)
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/layer.py:424:
UserWarning: `build()` was called on layer 'transformer', however the
layer does not have a `build()` method implemented and it looks like
it has unbuilt state. This will cause the layer to be marked as built,
despite not being actually built, which may cause failures down the
line. Make sure to implement a proper `build()` method.
  warnings.warn(
```

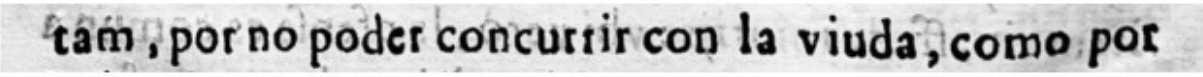
```
def preprocess(path, img_height=64):
    img = Image.open(path).convert('L')
    w, h = img.size
    new_w = max(1, int(w * img_height / h))
    img = img.resize((new_w, img_height), Image.BICUBIC)
    arr = np.array(img, dtype=np.float32)/127.5 - 1.0
    return tf.constant(arr[np.newaxis, :, :, np.newaxis],
dtype=tf.float32)


def ctc_greedy_decode(log_probs, idx2char, blank=0):
    best = tf.argmax(log_probs, axis=-1).numpy()
    results = []
    for row in best:
        col = [row[0]]
        for t in row[1:]:
            if t != col[-1]: col.append(t)
        results.append(''.join(idx2char.get(int(t), '') for t in col
if t != blank))
    return results

all_preds, all_gts = [], []

for i, item in enumerate(val_items):
    tensor = preprocess(item['image'], IMG_HEIGHT)
    lp     = model(tensor, training=False)
    pred   = ctc_greedy_decode(lp, idx2char)[0]
    all_preds.append(pred)
    all_gts.append(item['text'])
    if (i + 1) % 200 == 0:
        print(f'  {i+1}/{len(val_items)} done')
```

Testing

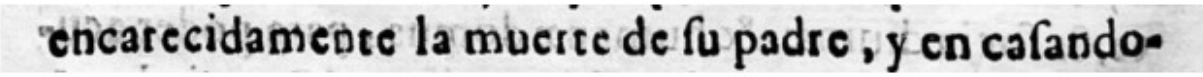
```
FOLDER_PATH = '/content/dataset/images/test/'
TEST_IMAGE = FOLDER_PATH + '*.jpg'

# 2. Use glob to find all matching files
IMAGE_PATHS = glob.glob(TEST_IMAGE)
IMAGE_PATHS.sort(reverse=True)
for path in IMAGE_PATHS:
    img = Image.open(path).convert('L')
    plt.figure(figsize=(8, 3))
    plt.imshow(img, cmap='gray')
    plt.axis('off')
    plt.show()

    tensor = preprocess(path, IMG_HEIGHT)
    lp     = model(tensor, training=False)
    pred   = ctc_greedy_decode(lp, idx2char)[0]

    print(f'Predicted text: "{pred}"\n')
```



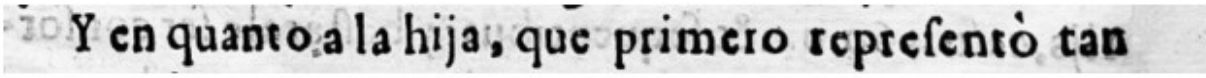

```
Predicted text: "hallarse sibdita de su marido, y probibida a acusalle
y"
```




```
Predicted text: "tam, por no poder concurir con la viuda, como por"
```




```
Predicted text: "encarecidamente la muerte de su padre, y en casando"
```



Predicted text: "Y en quanto a la hija, que primero re presento tan"

ço alentado, y con tantos agrauios a los ojos.

Predicted text: "qo alentado, y eon tanto agravios a los opos..i"

mano de la justicia, el temor no se le pierda este hijo

Predicted text: "rmano de la justicia, l temor no se le pierda este hijo"

que mas obliga a su madre a insistir en el castigo por la

Predicted text: " que mas obliga a su madre a insistir en el castigo por la"

hijo està tan lexos de remitir, que es vna de las causas

Predicted text: "hijo esta tan lexos de remitir, que es vna de las causas"

veras con que lo haze, y lo fia todo de su diligencia: y el

Predicted text: " veras con que lo har,y lo fia todo de su diligencia y el"

ce el primer lugar que tiene su madre para acusar, y las

Predicted text: "ce el primer luvgar que tiene su madre para acuar,y las,"

como queda fundado,nu.4. Lo ſegundo,porque el hijo

Predicted text: " como queda fundado, nu. 4 Lo segundo, porque el hijo"

de concurſo de los hijos con la viuda,no es admiſsible,

Predicted text: "de concurso de los hijos con la riuda, no es admisible,"

rido hazen mayor numero en concurſo con la madre,

Predicted text: "rido hazen may or numero en concurso con la madre,"